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RESEARCH ARTICLE

Advancing Malaria Identification From Microscopic Blood Smears Using Hybrid Deep Learning Frameworks

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ABSTRACT Malaria is a mosquito-borne, life-threatening, and contagious disease that has caused thousands of fatalities in recent years. Due to inadequate detection, the inexperience of laboratory personnel, and lack of advanced point-of-care equipment, the malaria-induced mortality rate is increasing. In addition to the traditional detection mechanisms, researchers have recently been investigating microscopic malaria-infected Red Blood Cells (RBC) image analysis based on deep learning models to detect malaria parasites as a general-purpose point-of-care solution. In this paper, we develop several hybrid data-driven models by combining a convolutional neural network (CNN) to extract the relevant features and two cascaded recurrent neural networks (RNNs) classifiers. Gated recurrent unit (GRU), long short-term memory (LSTM), and Bi-directional LSTM (BiLSTM) are considered candidate RNN classifiers. The models are compared in terms of accuracy, type-I & II error rates, and inference & training computation time. Experimental results demonstrate that the CNN-LSTM-BiLSTM model outperforms the other models with a significantly higher accuracy (96.20%), less type-I error rate (2.23%), and fewer combined type-I and type-II errors (3.80%). Also, we consider the model computation time (both training time per epoch and inference time per step)‘ as an important metric for emerging distributed learning paradigms where point-of-care devices with IoT (Internet of Things) capability can jointly contribute to global model accuracy. Thus, our findings demonstrate the practicality of cascading classifiers in resource-constrained point-of-care devices.

INDEX TERMS Malaria, detection, classification, CNN, LSTM, GRU, Bi-directional LSTM, hybrid model, cascaded classifier.

I. INTRODUCTION

Even today, malaria remains a life-threatening disease with a high infection risk in tropical regions and many other non-tropical areas [1], particularly when it goes undetected at an early stage. According to the World Malaria Report released by the World Health Organization (WHO), malaria cases rose to 12% compared to the preceding year, and 241 million malaria cases and nearly 627,000 deaths were reported in 85 countries in 2020 alone [2]. Malaria disease is caused by plasmodium parasites where the female anopheles mosquitoes are responsible for bearing these parasites and spreading this disease through mosquito bites [3]. The application of point-of-care disease detection strategies, inspired by the lessons learned from the recent COVID-19 pandemic, can greatly aid in the timely identification of these tropical and sub-tropical diseases, addressing a critical global health challenge.

However, the recent news of the ‘‘return’’ of malaria in the United States, with cases reported in Florida and Texas, highlights the ongoing threat of this disease. In the 1950s, the U.S. declared it had wiped out malaria through various public health measures. However, malaria never completely disappeared from the United States, with approximately

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2,000 cases diagnosed each year. The recent cases in Florida and Texas, not linked to international travel, raise concerns about the re-emergence of locally transmitted malaria [4]. We need to justify and point out that malaria is making a comeback despite the widespread thought that it has been eradicated.

This situation emphasizes the importance of early detection and effective disease management. Traditional microscopic tests for detecting plasmodium parasites responsible for malaria, such as Peripheral Blood Smears (PBS), are subject to reagent quality and technician skills [5]. Another commonly used method is the Quantitative Buffy Coat (QBC) technique, which involves collecting a blood sample in a hematocrit tube with acridine orange and anticoagulant. The presence of parasites is indicated by a bright green color under an epi-fluorescent microscope [6]. Even though this technique is highly reliable and user-friendly, this intervention requires costly instrumentation [7]. Another technique named Rapid Diagnostic Test (RDT) has been endorsed by the World Health Organization (WHO) as an alternative to microscopy tests. This rapid test is highly simple, fast, and cost-effective. A drop of blood can be tested using a simple kit, which detects the antigens or proteins produced by malaria parasites [8]. However, this method has limitations in detecting infections with low densities of malaria parasites [9]. Therefore, it is necessary to develop an automated, operable, cost-effective, and user-friendly technique that ensures high accuracy.

In this paper, we focus on the importance of point-of-care devices and the need for designing lightweight, data-driven models with high accuracy for fast and cost-effective detection of malaria, as depicted in Fig. 1. The objective is to find the optimal candidate model that balances high accuracy in malaria parasite detection from microscopic RBC images and computational efficiency, particularly in the context of point-of-care devices. Drawing inspiration from the gaps in existing techniques on malaria parasite identification, we develop different hybrid models to detect malaria from microscopic RBC images and evaluate the performance of our proposed hybrid cascading strategies with other previous deep learning models. Through our extensive experimentation on these microscopic images, we aim to contribute toward developing next-generation point-of-care devices that offer rapid, accurate, and cost-effective malaria diagnosis while considering the computational constraints of these devices. The key contributions of our work are outlined below.

1) We investigate standalone Convolution Neural Network (CNN) and Recurrent Neural Network (RNN) structures to detect malaria from microscopic blood cell images. While in the literature, CNN and RNN are known to effectively detect spatial and temporal patterns respectively, our research finds the RNN structure to be a powerful model for finding dependency correlations between different regions of the target microscopic blood cell images.
2) To leverage the respective strengths of CNN and RNN structures, we systematically combine them to design hybrid CNN-RNN models. As candidate RNN models, we consider the GRU (Gated Recurrent Unit), LSTM (Long Short Term Memory), and BiLSTM (Bi-directional LSTM). In our designed hybrid models, CNN is an automated feature extractor, while the RNN part acts as a classifier. Our research reveals that cascading two layers of RNN (i.e., using the same RNN types or different RNN types) to act as the classifier yields the best detection performance.
3) We perform a comprehensive comparative evaluation of the performances of the various models based on detection accuracy, precision, recall, F1 score, and type-I & type-II errors. We also consider the training/computation time as an important aspect, aligning with the need for distributed learning capability in next-generation point-of-care devices.

The remainder of this paper is organized as follows. Section II surveys the relevant research work. Section III describes our considered research problem. Our proposed methodology is presented in Section IV. The performances of our proposed hybrid models are evaluated in Section V. Section VI presents concluding remarks.

II. RELATED WORKS

A large body of research work in the literature considered malaria detection as a machine-learning classification task. For instance, Jdey et al. [10] proposed a Naïve Bayesian classifier to differentiate normal and abnormal blood cells for malaria detection and achieved 84% detection accuracy. On the other hand, Dharpal and Malviya [11] analyzed the performance of a Support Vector Machine (SVM) based on microscopic blood cell images. The study conducted by Dawale and Baraskar [12] introduced a combined model of Particle Swarm Optimization (PSO) with SVM to optimize the SVM classifier's kernel function radius in identifying malaria-infected blood cells. However, the SVM model was associated with challenges with quadratic programming and computational complexity when dealing with large-sized data [13]. This prompted a migration from machine learning to deep learning applications for malaria detection. Gourisaria et al. [14] demonstrated that a customized CNN model with a detection accuracy of 95.23% while Shah et al. [15] achieved even better performance by modifying the input data before training a customized CNN model. Next, Joshi et al. [16] developed a three-layered CNN structure and achieved 94.48% for testing phase malaria-infected cells. Alharbi et al. [17] trained a five-layered CNN model with image augmentation and obtained 95.56% accuracy. Researchers also explored the application of pre-trained CNN models, e.g., the ResNet50 model with 95.91% accuracy in detecting infected Red Blood Cells (RBCs) [18].

Other researchers also focused on considering computation time stemming from the practical implications of

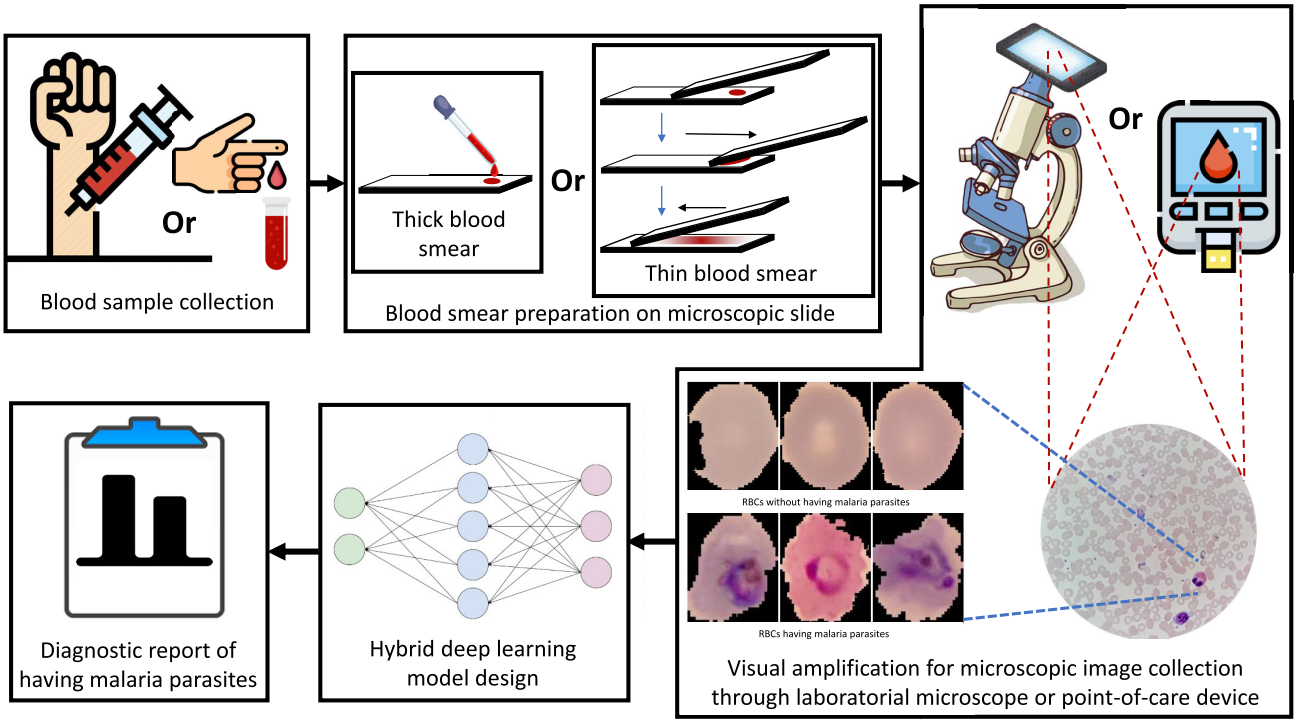


FIGURE 1. Envisioned deep learning model-assisted smart point-of-care device architecture for fast and cost-effective detection of malaria.

deploying machine learning models for real-time malaria detection. Point-of-care devices, such as mobile applications or handheld diagnostic tools, are designed to provide rapid and accurate results at the patient's location. These devices aim to minimize the turnaround time for diagnosis, enabling timely intervention and reducing the burden on healthcare systems. For instance, the study conducted by Sripon et al. [19] provided valuable insights into the tradeoff between model accuracy and computational efficiency. They compared several widely used pre-defined models, namely Xception, Inception-V3, ResNet-50, VGG-16, and AlexNet, using a dataset of 7,000 RBC images. The results obtained by Sripon et al. demonstrated Xception to emerge as the best performing model (with 99.28% accuracy), however, at the cost of a longer computation time of 125 minutes compared to the other models. This computational time can pose a challenge to next-generation, general-purpose point-of-care devices with IoT and distributed learning capability [20]. Regarding the current paradigm of IoT point-of-care devices for malaria detection, researchers focused on traditional mobile-based detection platforms. For instance, Masud et al. [21] designed a mobile application, leveraging a customized CNN model with a stochastic gradient descent (SGD) optimizer to identify malaria-infected RBCs, enabling quick and convenient diagnosis. However, to the best of our knowledge, none of the existing research work took into account both the performance of the models and the training time that is essential for the emerging point-of-device detection devices with IoT-based distributed learning features.

III. PROBLEM DESCRIPTION

Based on the research gap identified in the previous section, we now describe the research challenges associated with malaria detection. The problem is to systematically design a data-driven model that achieves better model performance in terms of error minimization and improved accuracy while ensuring reduced computation time to accommodate general-purpose, distributed learning-capable point-of-care diagnostic devices. Indeed, there has been a growing focus on developing intelligent point-of-care devices since the recent pandemic. While current devices are typically designed for diagnosing specific diseases, there is also a pressing need to prioritize the creation of general-purpose point-of-care monitors capable of diagnosing multiple diseases despite the constraints of their onboard resources. This is particularly crucial for conceptualizing a one-fit-for-all diagnostic platform for rapid detection of tropical and sub-tropical diseases, such as malaria, dengue, chikungunya, yellow fever, and so on. Therefore, it is essential to consider lightweight models that can achieve high accuracy and fast inference times on resource-constrained devices. Furthermore, emerging point-of-care devices are often equipped with Internet of Things (IoT) capabilities, enabling collaborative learning and scalable monitoring of disease propagation based on locally acquired data. With the computational limitations of these IoT-based devices, reducing training/computation time becomes a crucial factor. By identifying this need, our research addresses the tradeoff between overall detection accuracy (including type-I and type-II error rate mitigation) and local training time, which has been overlooked in

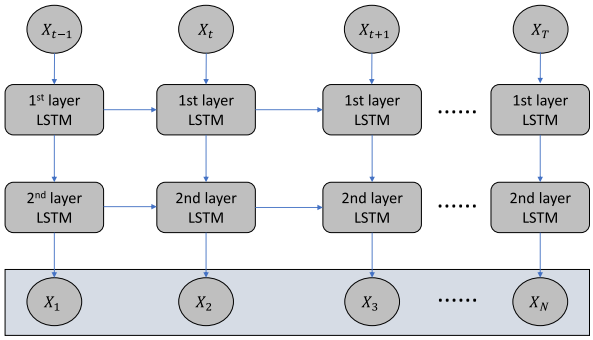


FIGURE 2. Two-layered LSTM.

previous studies. In the following section, we investigate how to design effective data-driven models that effectively address this tradeoff problem.

IV. ENVISIONED HYBRID DEEP LEARNING MODELS DESIGN

From here on, we develop the concept of hybrid deep learning models systematically to address the problem described in the earlier section. We split our design into three phases, namely automated feature extraction, a robust classifier design, and then the joint feature extraction and classifier. First, we explore the viability of a two-dimensional CNN for automating feature extraction. Next, we describe the RNN-based classifier, which is comprised of two layers. The layers of the RNN-based classifier are composed of the same or heterogeneous RNN types chosen from the pool of candidate models that include LSTM, BiLSTM, and GRU. Then, we present the cascaded CNN and RNN-based hybrid model architecture.

A. CNN-BASED AUTOMATED FEATURE EXTRACTION

CNN [22] is a deep learning-based architecture which is capable of directly learning from the input data. Variants of CNN have been demonstrated to be highly effective in detecting spatial patterns and extracting relevant features from imaging data, making them well-suited for tasks such as object recognition and classification. However, in our work, before the classification stage, we consider exploiting the spatial pattern detection capability of a two-dimensional CNN architecture to produce meaningful features from the input data (i.e., microscopic blood cell imaging data sample on the thin film placed on the point-of-care device). The CNN architecture consists of multiple layers, including convolutional layers, pooling layers, and fully connected layers. By incorporating CNNs as the feature extraction component, we aim to leverage their ability to capture relevant and discriminative information from the input image samples. Thus, in this stage, we obtain the most relevant features set to be passed to the next stage, i.e., a classifier for detecting the malaria parasite.

B. RNN-BASED CLASSIFIER DESIGN

RNNs [23] have been demonstrated to be effective in learning long-term dependencies, particularly sequence prediction tasks. One prominent RNN variant, known as LSTM, incorporates feedback connections that enable it to process entire sequences of data. Bi-directional LSTM (BiLSTM) is another type of RNN commonly used in natural language processing (NLP). Unlike standard LSTMs, BiLSTMs process input in both forward and backward directions, permitting them to leverage information from both past and future contexts. This bi-directional flow of input is what sets BiLSTMs apart from regular LSTMs, which process input in only one direction. Incorporating information from both directions makes BiLSTMs particularly effective in capturing long-term dependencies [24]. Figs. 2 and 3 depict a better understanding of the two-layered LSTM and Bi-directional LSTM models, respectively. GRUs are another type of gated mechanism within RNNs. GRUs resemble LSTMs but possess fewer parameters as they lack an output gate [25].

In conventional classification tasks, RNN models, namely LSTM, BiLSTM, and GRU models, are primarily applied to scenarios involving sequential data rather than individual data points, such as images. However, when relevant image features are extracted in the earlier stage and passed to the RNN structure, the latter can act as a powerful classification model to find dependency correlations between different regions of the target microscopic blood cell images for malaria parasite detection. Therefore, in the classifier stage, we aim to combine different RNN types/configurations as classifier(s) in conjunction with the preceding feature extraction part leveraging CNN. As candidate RNN models, we consider LSTM, BiLSTM, and GRU. As for the classifier, we cascade two RNN layers comprising the same RNN type or heterogeneous RNN types. For instance, we develop the CNN-LSTM-LSTM model whereby the two-layer LSTM serves as the cascaded classifier. On the other hand, in the case of the CNN-LSTM-GRU, the LSTM is employed as the initial classifier whereas GRU is employed in the second layer. Similarly, CNN-LSTM-BiLSTM and CNN-LSTM-GRU models are designed by considering LSTM as the first layer in the overall classifier module. Similar to the LSTM-based models, combined configurations involving CNNs and GRUs have been investigated. For instance, the CNN-GRU-GRU model utilizes a two-layer GRU as the cascaded classifier. In the CNN-GRU-LSTM and CNN-GRU-BiLSTM models, the GRU serves as the first classifier layer. Similarly, in the context of a combination of CNNs and BiLSTMs, various configurations have been explored. First, we consider the CNN-BiLSTM-BiLSTM model employing a two-layer BiLSTM as the cascaded classifier. In the CNN-BiLSTM-LSTM and CNN-BiLSTM-GRU models, the BiLSTM acts as the first classifier layer.

C. HYBRID MODEL FUSING FEATURE EXTRACTION AND CASCADED CLASSIFICATION LAYERS

Based on the earlier described CNN-based automated feature extraction module and RNN-based cascaded classification layers, we are now ready to combine them to formulate a hybrid model for malaria detection. The hybrid framework

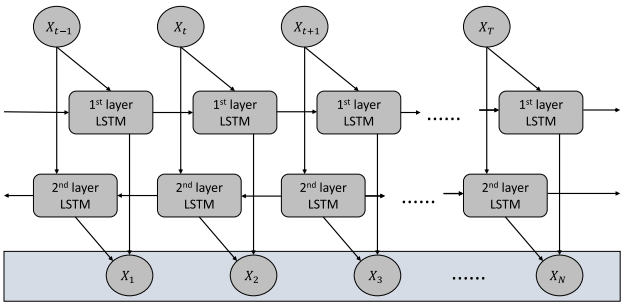


FIGURE 3. Bi-directional LSTM.

is depicted in Fig. 4. The feature extraction part of this model is performed by a 2-layered 2D CNN architecture. The input data are first fed through the BatchNormalization layer with the input size (None, 32, 32, 3). After the BatchNormalization layer, a 2D convolutional layer is constructed with a kernel size of 7, 64 filters, and ReLU (Rectified Linear Unit) as the activation function. Padding with zeros evenly is employed to the left/right or up/down of the input. This ensures that the filters fully cover the input data size. Note that the resultant output size is (None, 32, 32, 64) due to the use of 64 filters. Next, a 2-dimensional Maxpooling layer is applied with a pooling size of 2. This causes the output size to become (None, 16, 16, 64). Following this, a second 2-dimensional convolution layer is applied with a kernel size of 3, 128 filters, and ReLU as the activation function. Thus, the output dimension becomes (None, 16, 16, 128). The final layer of the feature extraction architecture comprises a 2-dimensional Maxpooling layer with a pooling size of 2. This causes the output size to become (None, 8, 8, 128).

Next, a combined classifier is cascaded to solve the long-term dependency issue in the input RBC imaging data. The combined classifier consists of two RNN layers, as discussed earlier. These RNN layers are sequentially executed. The output shape is transformed to (None, 128, 64) using the reshape method before utilizing the combined classifier. The units of the first and second RNN layers are set to 128 and 64 units, respectively, with both layers having a true return sequence. Consequently, the output shapes for the first and second RNN layers become (None, 128, 128) and (None, 128, 64), respectively. The sample data are then flattened and passed through two dense layers with 64 and 2 units, respectively, and a dropout layer with a 15% rate. Finally, the SoftMax function is employed as an activation function in the last dense layer.

In addition, the CNN-LSTM-LSTM, CNN-GRU-GRU, and CNN-BiLSTM-BiLSTM models are designed using 2-layered LSTM, 2-layered GRU and 2-layered BiLSTM models, respectively, without combining two distinct classifiers. In the aforementioned models, each layer is sequentially executed, and the output sizes of the classifiers are set to (None, 128, 128) and (None, 128, 64), respectively. For our proposed developed model, the initial learning rate was set to 0.02. We implemented a learning rate schedule that utilizes a momentum of 0.9 with the Stochastic Gradient Descent (SGD) optimizer. The learning rate was not scheduled to change dynamically; however, we have used the EarlyStopping callback, which was implemented to prevent overfitting and to stop training when the validation loss ceased to decrease over 10 epochs. The batch size for training each variant of the model was consistently set at 32.

V. RESULT AND ANALYSIS

In this section, we present the performance evaluation of our envisioned hybrid model for malaria detection. We first discuss the data collection and preparation steps. Then we analyze the performance of our proposal in terms of training, validation, and test accuracy along with other metrics including precision, recall, F1 score, computation/training time, type-I, and type-II errors for all the configurations of the hybrid model. We also compare the performance of each configuration of our proposed hybrid model with baseline models from the literature. The entire study was executed in the cloud-based environment of Google Colab Pro (Device Specification: core i7 processor, 16GB RAM, 64-bit operating system).

A. DATA COLLECTION

We employed a public malaria dataset based on the TensorFlow online dataset library that was obtained from the ListerHill National Center for Biomedical Communications. The dataset contains 27,558 cell images labeled as parasitized cells and uninfected cells based on the presence of plasmodium parasites. In the dataset, a total of 13,779 blood cells were denoted as parasitized cells and labeled as class 0 (zero). Similarly, the rest 13,779 blood cells were denoted as uninfected cells and labeled as class 1 (one) [26]. The data labeling process of malaria cells, along with the microscopic visualization, is depicted in Fig. 5.

B. DATA SAMPLING AND PREPROCESSING

We utilized the samples of 27,558 RGB (it refers to a system that represents an image by three basic colors, i.e., red, green, and blue) samples as our dataset. The whole dataset was split into 80% training, 10% validation, and 10% test data. The input data size of each sample was considered to be 32×32. For the preprocessing, we employed the ImageDataGenerator class. ImageDataGenerator is used for transforming the original data into a random basis and image augmentation. Image data Augmentation refers to the process of transforming a single image into multiple copies along with a few transform techniques [27]. For our considered dataset, we applied image rotation of approximately 20 degrees, vertical and horizontal flipping.

C. MODEL PERFORMANCE EVALUATION

Our model performance is evaluated via the confusion matrix, representing how many samples are correctly or wrongly identified. There are a few metrics that were calculated through four performance measures; namely, True Positive

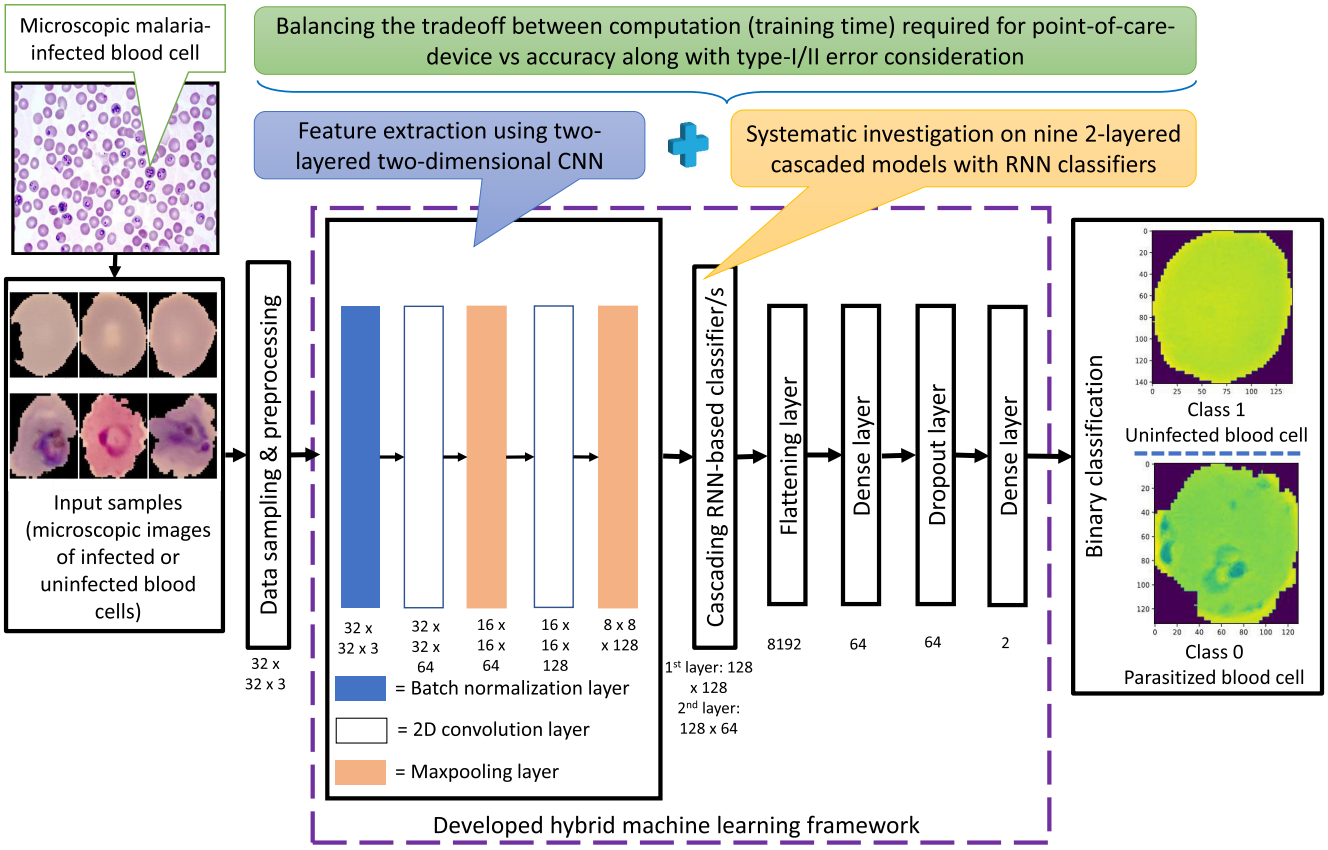


FIGURE 4. **Proposed hybrid framework illustrating feature extraction using a 2D CNN model and classification via cascaded 2-layered RNN classifiers incorporating LSTM, GRU, and BiLSTM. A comparative study involving nine CNN-RNN models was conducted, considering the tradeoff between model performance and computation time.**

TABLE 1. **Confusion matrix.**

		True Classes: Infected	True Classes: Not Infected
Model Decision	Infected	True Positive (TP)	False Positive (FP) Type-I Error
	Not Infected	False Negative (FN) Type-II Error	True Negative (TN)

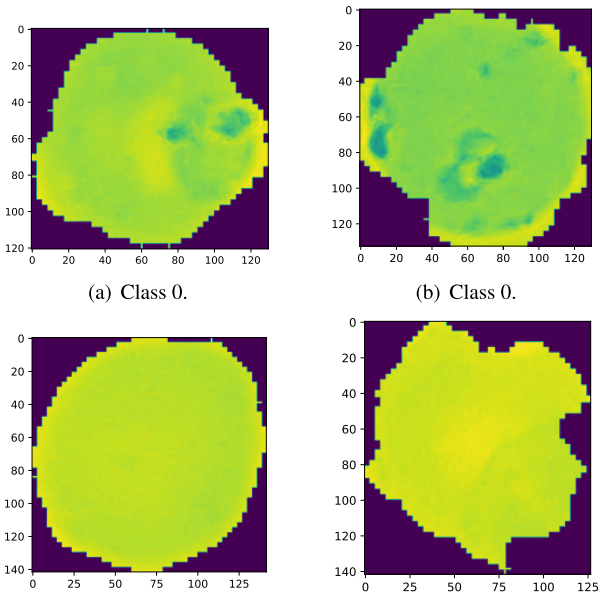
(TP), True Negative (TN), False Positive (FP), and False Negative (FN). TP samples are truly classified as positive samples by both the classifier and previous data labels. Similarly, TN samples are truly classified as negative samples by both classifier and data labeling. On the other hand, FP samples are those which are wrongly classified as positive samples by a classifier. This type of error is remarked as the type I error. Similarly, FN samples are those that are wrongly identified as negative samples, and this type of error is referred to as the type II error [28]. In our work, the type II error can be regarded as the worst scenario for contiguous diseases. However, we also need to consider type-I errors as falsely prescribed medications could harm healthy patients. For ease of understanding, the performance measures of the confusion matrix are shown in Table 1 that map the model decision to the infected and not infected classes.

According to the aforementioned classification matrix, we analyzed our model performance through a few evaluation parameters, such as precision, sensitivity or recall, classification accuracy, and F1-score. Precision refers to the ratio of truly positive samples and all labeled positive samples by the classifier, representing all positive samples and how many of the samples are truly classified as positive. Sensitivity signifies the ratio between truly positive samples and all labeled positive samples by the previous data labeling, which refers to how many samples of previously labeled data were truly classified as positive samples through the classifier. On the other hand, the F1-score indicates the harmonic mean of sensitivity and precision, which defines whether a model can perfectly classify both positive and negative samples or not [29]. If the F1-score of a model performance exceeds 0.8, the model is considered to be well-built [30]. The evaluation metrics are expressed as follows.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \tag{1}$$

$$Precision = \frac{TP}{TP + FP}, \tag{2}$$

$$Recall = \frac{TP}{TP + FN}, \tag{3}$$



(a) Class 0. (b) Class 0. (c) Class 1. (d) Class 1.

FIGURE 5. Images of Parasitized (a & b) and uninfected (c & d) blood cell labelling.

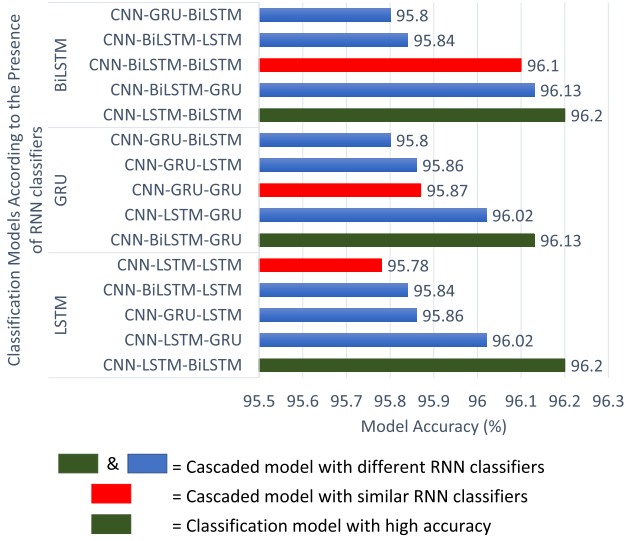


FIGURE 6. Model accuracy according to cascaded 2-layered similar and different RNN classifiers. Blue & green-colored bars represent cascaded models with different RNN classifiers; among them, the green-colored bar represents the classification model with high accuracy. On the other hand, the red-colored bar represents the cascaded model with similar RNN classifiers.

$$F1\,Score = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2 * TP + FP + FN}. \quad (4)$$

D. ANALYSIS ON CLASSIFICATION ACCURACY

In this section, we compare the performance of the different hybrid models we developed. To analyze the accuracy of each model, we split the entire dataset into training (80%), validation (10%), and test (10%) data. Each model was executed three times to ensure the reliability of the performance metrics and included the mean and variance of the accuracies for each nine developed hybrid models. Table 2 displays the classification accuracy of each model on the training, validation, and test data. Among all hybrid models, the CNN-LSTM-BiLSTM model achieved the highest accuracy (96.20 ± 0.11)% on the test dataset, followed by the CNN-BiLSTM-GRU model (96.13±0.15)%. The lowest model accuracy was achieved by the CNN model with two-layered LSTM classifiers (95.78 ± 0.16)%. In terms of precision, recall, and F1-score, CNN-LSTM-BiLSTM and CNN-BiLSTM-BiLSTM models yielded the highest (0.97, 0.97, and 0.97) and lowest (0.94, 0.94, and 0.94) values, respectively. The remaining models performed similarly in terms of precision (0.96), recall (0.96), and F1-score (0.96), except for the CNN-LSTM-GRU model, which was associated with the precision, recall, and F1-score values of 0.95, 0.95, and 0.95, respectively.

Next, Fig. 6 demonstrates that the cascaded models with different RNN architectures generally achieved higher mean accuracy compared to the cascaded models with similar RNNs. In other words, the CNN-LSTM-BiLSTM and CNN-BiLSTM-GRU models, both incorporating different types of RNNs in the cascaded form, achieved higher accuracy (96.20% and 96.13%, respectively) compared to the models with the same RNN in the cascaded configuration, such as CNN-LSTM-LSTM (95.78%) and CNN-GRU-GRU (95.87%). The advantage of using a cascaded model with different RNN architectures lies in the diversity and complementary strengths of each RNN type. Different RNNs, such as LSTM, GRU, and BiLSTM, have unique architectural variations and learning capabilities. The diversity among the RNN architectures reduces the risk of model bias and can improve generalization and robustness within the image data [31]. In the context of malaria identification from infected blood cells, employing a variety of RNN architectures in the cascaded configuration allows the model to capture different temporal dependencies and patterns in the data. Thus, this combination of diverse RNN types enables the proposed model to leverage its respective strengths and capture various temporal dependencies to substantially improve the overall model performance.

E. ANALYSIS OF COMPUTATION TIME

A computation time of a model is needed so that we can know whether the developed model is capable of running within the preferred time. It holds great significance for point-of-care devices, as it determines whether a developed model can operate within the desired time frame, offering a particular advantage toward distributed learning among a group of point-of-care devices with IoT capability. These devices operate in real-time scenarios where prompt and efficient diagnoses are crucial. Therefore, models with shorter computation and inference times are highly desirable for point-of-care devices, as they enable rapid and cost-effective disease detection, as well as distributed learning by locally

TABLE 2. Classification metrics of developed hybrid models.

Classification Models	Training Accuracy (%)	Validation Accuracy (%)	Test Accuracy (%)	Precision	Recall	F1 score
CNN-LSTM-LSTM	**97.98 ± 0.51**	95.65 ± 0.13	95.78 ± 0.16	0.96	0.96	0.96
CNN-GRU-GRU	97.84 ± 0.26	95.77 ± 0.09	95.87 ± 0.02	0.96	0.96	0.96
CNN-GRU-LSTM	97.70 ± 0.12	95.56 ± 0.25	95.86 ± 0.05	0.95	0.95	0.95
CNN-LSTM-GRU	97.76 ± 0.13	95.51 ± 0.27	96.02 ± 0.04	0.96	0.96	0.96
CNN-BiLSTM-BiLSTM	97.39 ± 0.18	**95.83 ± 0.17**	96.10 ± 0.16	0.94	0.94	0.94
CNN-LSTM-BiLSTM	97.64 ± 0.22	95.15 ± 0.40	**96.20 ± 0.11**	**0.97**	**0.97**	**0.97**
CNN-BiLSTM-LSTM	97.57 ± 0.10	95.45 ± 0.27	95.84 ± 0.08	0.96	0.96	0.96
CNN-GRU-BiLSTM	97.86 ± 0.10	95.47 ± 0.40	95.80 ± 0.16	0.96	0.96	0.96
CNN-BiLSTM-GRU	97.75 ± 0.09	95.26 ± 0.13	96.13 ± 0.15	0.96	0.96	0.96

training their respective models without the need to share the exact data with other devices and thereby facilitate a global model training. By optimizing the computation time, the overall efficiency and affordability of point-of-care devices can be improved, ensuring timely and accurate diagnoses in resource-constrained settings.

Additionally, minimizing computation time helps reduce overall computation costs. In this study, we utilized GPU for each model and employed 100 epochs in conjunction with the EarlyStopping method. Each iteration consisted of 689 steps, with the first iteration taking slightly longer than the subsequent ones. For each model, the number of epochs computation time was different. Table 3 exhibits the mean and variance of computation time per epoch executing on training and validation datasets. From there, we can observe that CNN-BiLSTM-GRU required less mean computation time(9.11sec) although the mean testing accuracy (95.87%) is a bit less than the CNN-LSTM-BiLSTM model.

Analyzing the inference time of each model, which refers to the duration it takes for the model to make predictions on unseen data (i.e., test data) after being trained, is also important. In addition to the computation time on training data, inference time is a significant performance metric, particularly for real-time data applications, as this can help reduce operational costs [32]. From Table 3, it can be seen that the CNN model cascading with only LSTM and GRU classifiers required less inference time (5ms/step) than other models. The highest inference time was required for the CNN-BiLSTM-BiLSTM model (9ms/step). Thus, it is evident that the inference time is dependent on the size and complexity of the underlying model.

F. STRATEGIC MODIFICATION OF THE CONVENTIONAL CONV-RELU-BN CONFIGURATION

Batch Normalization (BN) is a technique used in neural networks, particularly in convolutional neural networks (CNNs), to speed up training procedures by normalizing the input in each layer. However, in our case, we noticed that incorporating BN after every convolutional layer increased the model complexity and did not improve model performance. Additionally, this approach led to a slight increase in overfitting and the computation training time.

TABLE 3. Training time per epoch and inference time per step for each model.

Classification Models	Training Time per Epoch (s)	Inference Time per Step (ms)
CNN-LSTM-LSTM	9.40 ± 0.22	5.00
CNN-GRU-GRU	**9.11 ± 0.35**	**5.00**
CNN-GRU-LSTM	9.15 ± 0.18	5.00
CNN-LSTM-GRU	9.16 ± 0.29	5.00
CNN-BiLSTM-BiLSTM	**16.01 ± 0.26**	**9.00**
CNN-LSTM-BiLSTM	14.96 ± 1.16	8.00
CNN-BiLSTM-LSTM	12.67 ± 0.14	7.00
CNN-GRU-BiLSTM	12.31 ± 0.15	7.00
CNN-BiLSTM-GRU	12.36 ± 0.12	7.00

For a better explanation, we conducted a comparative analysis of our best found model i.e. 2-layered CNN with LSTM-BiLSTM model with and without BN after every convolutional layer, shown in Table 4. We found that even though the training accuracy with BN is 98.18%, it decreased quite a lot for test accuracy (95.83%), which hints at a slight overfitting. Furthermore, with BN layers, the computation training time increased to 269.30s and inference time per step increased to 8.10ms, while without BN layers, the computation training time was 253.80s and the inference time per step was 7.60ms. From this observation, we conclude that the standard procedure of applying BN after every convolutional layer may not be the most optimal strategy, considering both performance metrics and computational efficiency.

G. RECEPTIVE FIELD VS. EFFICIENCY IN EXTENDED CNN-RNN

To find the dependency correlation between different regions of the target microscopic blood cell images through our proposed model, we extended our best-found model from a 2-layered CNN cascading with an LSTM-BiLSTM model to a 4-layered CNN before cascading with an LSTM-BiLSTM, effectively increasing the receptive field of the CNNs (shown in Table 5). The outcomes from there show that the test accuracy for the 2-layered CNN-RNN model is quite higher (96.08%) than the 4-layered CNN-RNN model

TABLE 4. Comparative analysis of the performance of CNN-LSTM-BiLSTM models with and without Conv-ReLU-BN configurations.

Classification Models	No. of Epochs	Training Accuracy	Validation Accuracy	Test Accuracy	Computation Training Time	Inference Time per Step
CNN-LSTM-BiLSTM	20	97.50%	95.50%	**96.08%**	**253.80s**	**7.60ms**
CNN-LSTM-BiLSTM with Conv-ReLU-BN	20	**98.18%**	**96.08%**	95.83%	269.30s	8.10ms

TABLE 5. Comparative analysis of CNN-LSTM-BiLSTM architectures with varied convolutional depths in cascading LSTM-BiLSTM classifiers.

Classification Models	No. of Epochs	Training Accuracy	Validation Accuracy	Test Accuracy	Computation Training Time	Inference Time per Step
2-layered CNN with LSTM-BiLSTM	20	97.50%	95.50%	**96.08%**	**253.80s**	**7.60ms**
4-layered CNN with LSTM-BiLSTM	20	**97.61%**	**95.54%**	95.54%	288.89s	8.60ms

(95.54%). Moreover, we have analyzed the computation times, including both inference time per step and total training time. Our analysis revealed that the 4-layered CNN cascaded with the LSTM-BiLSTM model requires more time, with an inference time per step of 8.60ms and a training time of 288.89s. This is notably higher compared to the 2-layered CNN cascaded with the LSTM-BiLSTM model, which has an inference time per step of 7.60ms and a training time of 253.80s. Therefore, it is evident that our proposed CNN-RNN model still outperforms the more complex CNN-RNN structure, suggesting that our approach effectively balances the trade-off between receptive field size and computational efficiency.

H. ANALYSIS OF TYPE-I AND TYPE-II ERRORS

The confusion matrix provides valuable insights into the accuracy of classification models by revealing the numbers of correctly and incorrectly classified positive and negative samples. In the experiments we conducted, we evaluated the performance of our developed hybrid models and computed the corresponding confusion matrices. Table 7 presents the percentages of False Positives; FP (type-I errors) and False Negatives; FN (type-II errors) based on the test data consisting of 2,756 samples executing three times. Analysis of the confusion matrices reveals that the CNN-LSTM-BiLSTM model achieved the lowest type-I error rate of 2.23% while the CNN-BiLSTM-GRU model exhibited the lowest type-II error rate of 1.33%. These results highlight the effectiveness of these specific hybrid models in accurately classifying the test samples, minimizing both types of classification errors.

Based on the evaluation metrics, including classification accuracy on the test dataset, precision, recall, F1-score, and type-II error rate, the CNN-LSTM-BiLSTM model outperformed the other classification models. However, it is worth noting that this model exhibited a slightly higher type-II error rate (1.57%), and yielded a longer training computation time per epoch (14.96 ± 1.16) seconds, as well as a higher inference time (8 milliseconds per step). On the other hand, the CNN-BiLSTM-GRU model demonstrated the lowest type-I error rate (1.33%). Although the CNN-GRU-GRU model required the least computation time per epoch (9.11 ± 0.35) seconds, it did not perform equally impressive in other classification metrics. These findings indicate the tradeoff between different models in terms of accuracy, error rates, and computational efficiency. The CNN-LSTM-BiLSTM model delivered superior classification performance overall, albeit with slightly longer computation and inference times. Conversely, the CNN-GRU-GRU model was attributed to faster training computation and inference time; however, its performance, in terms of other metrics, was comparatively less noteworthy.

I. DIFFERENCE BETWEEN ENVISIONED AND EXISTING MODELS

As mentioned in Section II, numerous studies have employed various classification algorithms for malaria disease detection. In this section, we conduct a comparative analysis of our developed best model among nine hybrid models with existing approaches based on the number of data utilization, classification accuracy (with a focus on testing accuracy), and type-I and type-II error rates. Table 6 presents a summary of the performance of these models.

The Naïve Bayes classifier achieved the lowest classification accuracy of 84% using a dataset of 27,558 samples, and it exhibited the highest type-II error rate of 13.33% among all models. The pre-defined CNN model, VGG-19, with fine-tuning, achieved the second lowest type-II error rate of 0.97%, despite having a lower classification accuracy (95.56%) compared to our models. Our developed hybrid CNN-LSTM-BiLSTM model exhibited the lowest type-I error rates (2.23%) compared to other models while achieving a better classification accuracy of 96.20% on the test data.

Our future research focus will lead to developing more sophisticated and accurate models. Specifically, we will refine these models to reduce type-I and type-II error rates further. Our goal is to enhance not only the performance of our models quantitatively but also their computation efficiency to make a point-of-care device for real-world applications. Furthermore, we plan to conduct extensive experiments using attention-based models and other complex

TABLE 6. Comparison among existing and our envisioned models.

Reference	Proposed Method	Test Accuracy	No. of Data	Type-I Error	Type-II Error
[10]	Naïve Bayes classifier	84%	27,558	2.66%	13.33%
[17]	**VGG-19 with fine-tuning**	95.56%	27,558	2.3%	**0.97%**
[14]	ResNet50 with transfer learning	95.71%	27,558	N/A	N/A
Our work	**CNN-LSTM-BiLSTM**	**96.20%**	27,558	**2.23%**	1.57%

TABLE 7. Type-I and type-II errors of the developed hybrid models.

Classification model	Type-I Error	Type-II Error
CNN-LSTM-LSTM	2.56%	1.66%
CNN-GRU-GRU	2.47%	1.66%
CNN-GRU-LSTM	2.42%	1.56%
CNN-LSTM-GRU	2.58%	1.56%
CNN-BiLSTM-BiLSTM	2.35%	1.55%
CNN-LSTM-BiLSTM	**2.23%**	1.57%
CNN-BiLSTM-LSTM	2.56%	1.57%
CNN-GRU-BiLSTM	2.85%	1.34%
CNN-BiLSTM-GRU	2.54%	**1.33%**

feature fusion strategies across various datasets and under different conditions.

VI. CONCLUSION

The recent re-emergence of malaria in various parts of the world is triggering a growing need for faster and more accurate detection methods. Utilizing a deep-learning approach to identify malaria parasites in infected blood cells, in point-of-care device settings, may prove to be a viable option that could offer fast detection time, as well as cost-effectiveness. In this paper, a few hybrid deep-learning models were developed. Our research findings indicate that cascading two different RNN classifiers mostly outperformed better than cascading similar RNN classifiers. In particular, our proposed CNN-LSTM-BiLSTM model exhibited superior performance compared to other models, achieving a high model accuracy of 96.20% and the lowest type-I error rate of 2.23%. On the other hand, The CNN-BiLSTM-GRU model demonstrated the lowest type-II error rate of 1.33% and the second-highest accuracy (96.13%) among all models. The CNN-GRU-GRU model demonstrated better performance than other models in terms of shorter training computation time per epoch (9.11 ± 0.35) seconds, however, its other classification metrics were not as impressive. Regarding the inference time per step, CNN-BiLSTM-BiLSTM and CNN-LSTM-LSTM took 9msec and 5msec respectively to execute each step indicating that the inference time depends on the model's size and complexity.

These findings highlight the advantages of utilizing hybrid models with cascaded RNN classifiers for malaria parasite detection. However, further research and improvements are necessary to optimize the model performance. The application of these approaches extends beyond malaria detection and holds promise for advancements in smart point-of-care devices with IoT capability, enabling rapid and accurate diagnosis of malaria in resource-constrained settings. By leveraging deep learning techniques and optimizing computation time, point-of-care devices can offer cost-effective and timely detection of malaria, ultimately contributing to global efforts in combating this disease.

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